INCH-POUND
MIL-PRF-3098/67F
w/AMENDMENT 1
7 October 2019
SUPERSEDING
MIL-PRF-3098/67F
7 January 2010

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR89/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.12 MHz to 6.2 MHz; fundamental; noncontrolled; antiresonance.

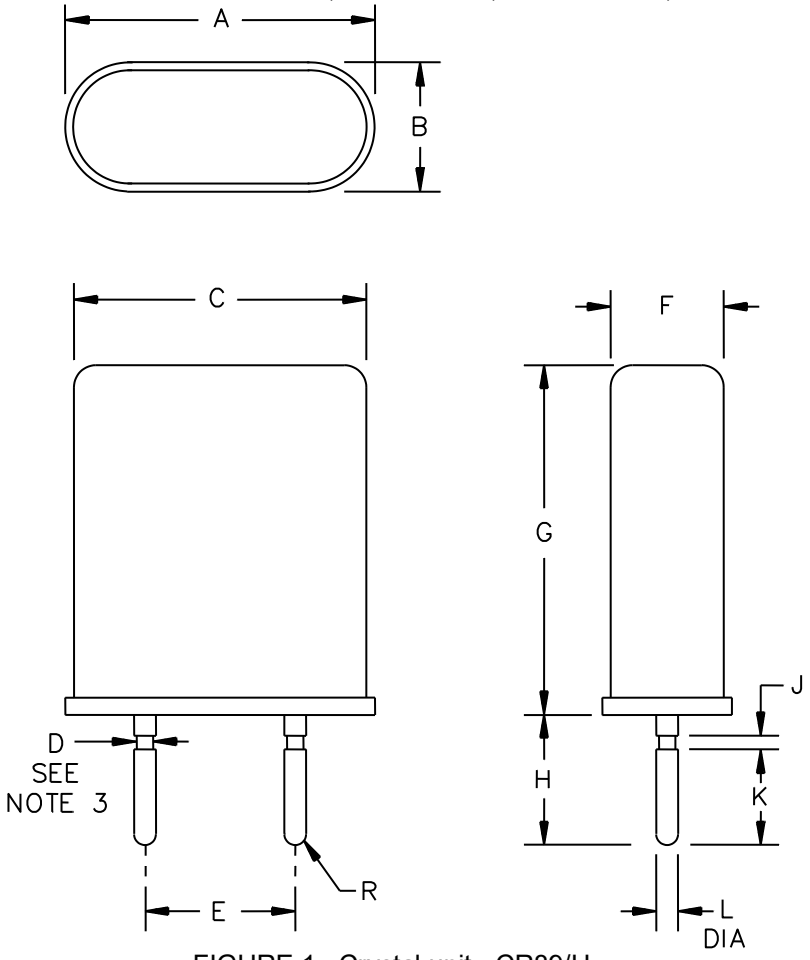


FIGURE 1. Crystal unit - CR89/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.567	---	14.40
B	---	.237	---	6.02
C	---	.535	---	13.59
D	.030	.037	0.76	0.94
E	.264	.303	6.71	7.70
F	---	.207	---	5.26
G	---	.640	---	16.26
H	.233	.248	5.92	6.30
J	.015	.025	0.38	0.64
K	.165	.175	4.19	4.45
L	.038	.042	0.97	1.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR89/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.12 MHz to 6.2 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 ppm,

Equivalent resistance: See table I.

Antiresonance, load capacitance: 30.0 pF ±0.5 pF.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: MIL-STD-202-201.

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Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Frequency and equivalent resistance:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive MHz	Maximum resistance Ohms
2.10 to 2.25	500
2.25+ to 2.80	320
2.80+ to 3.41	250
3.41+ to 4.10	180
4.10+ to 5.10	120
5.10+ to 6.20	75

Part or Identifying Number (PIN): CR89/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-201

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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